

BEM-C202
SEMESTER EXAMINATION- MAY 2024
B.TECH SEMESTER: I
ENGINEERING MATHEMATICS II

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

CO BL

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks. (5 × 6 = 30 Marks)

- | | | | |
|-----|--|-----|----|
| 1. | By understanding the concept of exactness solve the following differential equation | CO1 | L1 |
| | $(xy^2 - e^{\frac{1}{x^3}})dx - x^2ydy = 0$ | | |
| 2. | Find the solution of following differential equation | CO1 | L1 |
| | $y'' + 4y' - 12y = (x - 1)e^{2x}$ | | |
| 3. | By understanding the concept of linear partial differential equation solve | CO2 | L2 |
| | $2\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 3\frac{\partial^2 z}{\partial y^2} = 5e^{x+y}$ | | |
| 4. | Classify the nature of following partial differential equation | CO2 | L2 |
| | $(1 - x^2)\frac{\partial^2 z}{\partial x^2} - 2xy\frac{\partial^2 z}{\partial x \partial y} + (1 - y^2)\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} + 3x^2y\frac{\partial z}{\partial y} - 2z = 0$ | | |
| 5. | Express $4x^3 + 6x^2 + 7x + 2$ in terms of Legendre polynomials. | CO3 | L2 |
| 6. | Deduce that when n is a positive integer $J_{-n}(x) = (-1)^n J_n(x)$. | CO3 | L3 |
| 7. | Obtain the Fourier series expansion of $f(x) = \frac{\pi-x}{2}$ in the interval $(0, 2\pi)$. | CO4 | L4 |
| 8. | Obtain the half range sine series for the function $f(x) = x^2$ in $0 < x < 3$. | CO4 | L4 |
| 9. | A manufacturer knows that the condensers he makes contain on an average 1% of defectives. He packs them in a box of 100. What is the probability that a box picked at random will contain | CO5 | L5 |
| | (i) Exactly 2 faulty condensers | | |
| | (ii) 4 or more faulty condensers | | |
| 10. | A manufacturer of envelopes knows that the weight of the envelopes is normally distributed with mean 1.9 gm and variance 0.01 gm. Examine how many envelopes weighing | CO5 | L3 |
| | (i) 2 gm or more | | |
| | (ii) 2.1 gm or more | | |
- Can be expected in a given packet of 1000 envelopes.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 × 10 = 40 Marks)

1. Evaluate the following differential equation by changing the independent variable CO1 L5

$$x \frac{d^2 y}{dx^2} + (4x^2 - 1) \frac{dy}{dx} + 4x^3 y = 2x^3$$

2. Use variation of parameter method to solve CO1 L1

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = e^{-x} \log x$$

3. Use method of separation of variables to evaluate CO2 L5

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u, \text{ where } u(x, 0) = 6e^{-3x}.$$

4. Find the power series solution of the following differential equation about $x=0$ CO3 L3

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0.$$

5. Express that $\int_{-1}^1 P_m(x) P_n(x) dx = 0$ if $m \neq n$ CO3 L2

6. Express $f(x) = x + x^2$ for $-\pi < x < \pi$ in terms of Fourier series hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ CO4 L4

7. The following marks have been obtained by a class of students in mathematics CO5 L4

Paper I	80	45	55	56	58	60	65	68	70	75	85
Paper II	81	56	50	48	60	62	64	65	70	74	90

Calculate the coefficient of correlation for the above data. Find the lines of regression y on x and x on y .

8. Apply method of least square to obtain a parabola that approximates the data CO5 L3

x	1	1.2	1.4	1.6	1.8	2
y	2.345	2.419	2.592	2.863	3.233	3.702
